

ACHYUTH S S

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Software Engineer focused on **Python**, **backend systems**, **developer tooling**, and **AI-assisted engineering workflows**. Experienced in building **FastAPI services**, **agentic automation**, **MCP-based tooling**, **test automation**, and **local AI systems**. Strong in **debugging**, **automation**, **deterministic validation**, and **integrating coding agents into real engineering workflows**, with hands-on experience modernizing legacy systems and building Linux-based AI tooling.

EDUCATION

PES University, Bengaluru

GPA: 7.5

Bachelor of Technology, Electronics and Communication Engineering

Sep 2021 – Aug 2025

EXPERIENCE

KPMG India — Analyst

Bengaluru

AI Automation, Developer Tooling & Test Modernization

Jan 2025 – Present

- Developed scalable **Python FastAPI microservices** powering document processing SaaS platforms with focus on performance, reliability, and maintainability.
- Built a full-stack defect similarity platform using **Streamlit**, **FastAPI**, **PostgreSQL**, and **REST APIs**, integrating Azure DevOps, Jira, and other project management systems.
- Developed an **AI-assisted migration workflow** for modernizing legacy **UFT/VBScript automation into Python Selenium + Behave BDD**, handling test scripts, object repositories, reusable function libraries, and framework-specific code generation.
- Integrated **agentic coding workflows and MCP-based tooling** to provide structured engineering context to coding agents, with deterministic validation and human-review boundaries for application-specific artifacts.
- Built and debugged **Python automation tooling** for generating maintainable test artifacts while enforcing code-quality and framework conventions across AI-generated outputs.

Nokia Corporation — Project Trainee

Bengaluru

Apr 2024 – Jul 2024

- Developed **Python-based backend services** integrating **distributed systems via REST APIs and service-to-service communication**.
- Designed **data processing pipelines** for large-scale network telemetry, enabling **anomaly detection and data-driven analysis**.
- Processed **high-volume datasets** using optimized SQL queries, improving **performance and reducing issue resolution time by ~40%**.

True Data Software — Full Stack Developer Intern

Remote

Aug 2023 – Oct 2024

- Developed full-stack ERP/CRM features using **React**, **JavaScript**, **Python**, and **REST APIs**, building responsive interfaces and scalable backend services.
- Implemented **data validation, transformation, and processing pipelines** for structured enterprise datasets.
- Developed **real-time data ingestion workflows and automation systems**, improving **system efficiency and operational scalability**.
- Improved backend efficiency and reduced manual effort by **~40 hours/month** through **process optimization and automation**.

PROJECTS

Project Pegasus

C++, Python, llama.cpp, whisper.cpp, Linux

- Engineering a **local-first AI voice assistant for Linux** targeting Debian- and Arch-based systems, enabling natural-language voice interaction with system and developer workflows.
- Building an end-to-end **speech-to-action pipeline** using **whisper.cpp** for on-device speech recognition, **llama.cpp** for local LLM inference, and C++-based orchestration for low-latency execution.
- Developing **agentic workflows** that translate natural-language commands into structured actions, while experimenting with model selection, prompt/tool routing, wake-word detection, and execution reliability.
- Optimizing the system for **offline/local inference**, evaluating quantized models and runtime trade-offs across latency, memory usage, and response quality on consumer Linux hardware.

Prediction of Handover Failures in Dense Cellular Networks

Python, scikit-learn

- Developed a **scalable data processing pipeline** for cellular network telemetry including feature engineering and structured log ingestion.
- Used **Pandas**, **NumPy**, and **machine learning models** for **anomaly detection and predictive analytics** on large-scale datasets.

PESU Dev

- Improved **authentication/authorization APIs and CI/CD workflows**, strengthening deployment reliability and overall code quality.
- Developed and maintained **pytest-based automated tests**, expanding test coverage and improving regression detection across backend components.
- Contributed to **honeypot detection and anti-abuse tooling** for the PESU Discord bot using **Python and discord.py**, implementing logic to identify and handle suspicious interaction patterns.
- Contributed to the **AskPESU chatbot frontend** using **React, TypeScript, and Vite**, implementing and improving user-facing functionality.
- Participated in **peer code reviews** across backend APIs, automation, and application components, improving code quality while gaining practical exposure to production-oriented system design and engineering practices.

SKILLS

Programming: Python, JavaScript, TypeScript, C, C++

Backend & APIs: FastAPI, Flask, REST APIs, Microservices, JSON/CSV Processing

AI & Automation: Agentic AI Workflows, MCP, AI-Assisted Development, Selenium, Behave BDD, Gherkin

Systems & Tooling: Linux, Git, Docker, CI/CD, Debugging, Logging, Testing, PyTest

Databases: PostgreSQL, MySQL, SQL, Query Optimization

Cloud: AWS (S3, EC2), Azure, GCP

Core Concepts: Data Structures, Algorithms, OOP, System Design, Distributed Systems